

Northwestern

A Graph Neural Network for Pandemic Forecasting and Planning: Final Presentation

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Background and Problem Recap

- Goal: Predict the spread of a given disease (in this case COVID-19) over a specific network of locations
- We define the following:
 - $Y_{s,t,v}$: The number of cases in state s at time t for variant v
 - $X_{s,t}$: Exogenous covariates for state s at time t
 - $G_S = (S, E)$: The state-level graph, with states as nodes, and edges being weighted connections between them
 - Edge weights are a combination of shared border length, proximity and size of nearby airports, and shared (traffic-weighted) highways

Graph Construction

- Formally, the highway weight is:

$$W_{ij}^H = \frac{M_i M_j}{C_{ij}^2}, M_i = \sum_i \text{AADT} \times \text{lanes} \times \text{miles}$$

where AADT is the average daily traffic, and C_{ij} is the distance in minutes from the centroids of the two states

- The airport weight is:

$$W_i^A = \sum_j \frac{E_i E_j}{\max(E_k)}$$

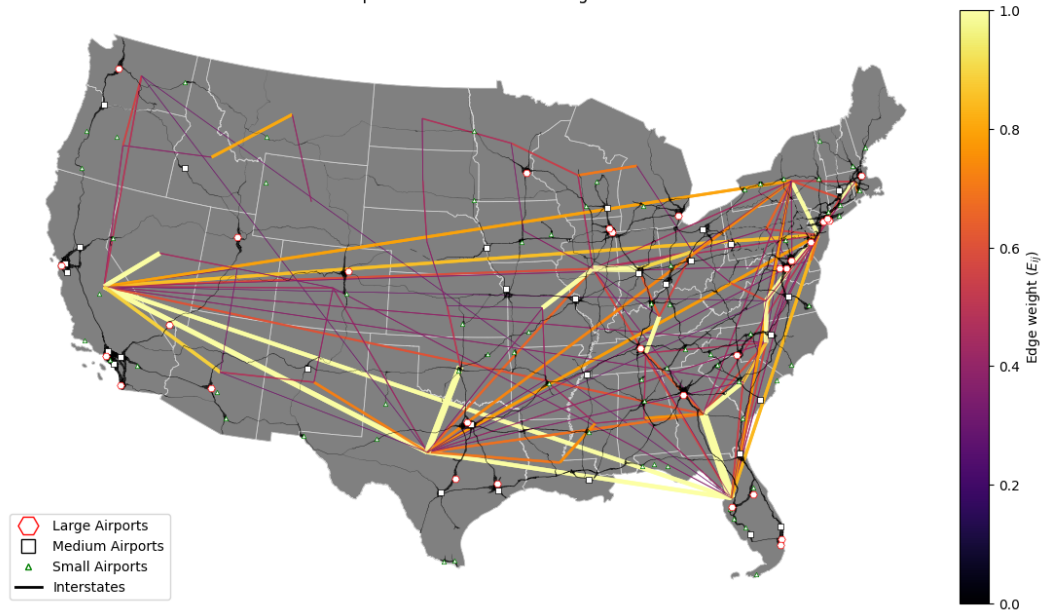
where E_i is the total enplanements (passengers boarding a flight) of all airports within 20 miles of state i

- The border weight is simply the length in meters of the shared border between two states

All weights are then normalized and summed

Graph Construction

Top 10% of State-to-State Edges



Graph-Based Baselines: StemGNN & MTGNN

1. StemGNN (Spatio-Temporal Enhanced Modeling GNN) An **auto-regressive** graph forecasting model that combines:

- **Spectral graph convolution** for spatial dependencies.
- **Discrete Fourier Transform** for temporal periodicity.
- Joint **spectral–temporal representation** for multivariate series.

2. MTGNN (Multivariate Time Series Graph Neural Network) A **seq-to-seq** graph neural network that integrates:

- **Graph learning module** that learns adjacency from data.
- **Mix-hop propagation** to enlarge receptive fields.
- **Temporal Convolution Network (TCN)** for long-range patterns.

Naive Baselines

1. Last-Day Prediction

$$\hat{y}_{s,t+k} = \sum_v y_{s,t,v}$$

- Uses the last observed value (summed over all variants) to predict all future steps.
- Serves as a simple, parameter-free lower bound.

2. Moving Average Prediction

$$\hat{y}_{s,t+k} = \frac{1}{P} \sum_v \sum_{i=t-P+1}^t y_{s,i,v}$$

- Smooths short-term fluctuations using recent observations.
- A stronger baseline when the data shows local variability.

Baseline Evaluation Results Overview

We evaluate all baselines on forecasting horizons from 1 to 7 days. The models include:

- **StemGNN**
- **MTGNN**
- **Last-Day baseline**
- **Moving Average baseline**

For each model, we report:

- **MAPE**: Mean Absolute Percentage Error
- **RMSE**: Root Mean Squared Error
- **MAE**: Mean Absolute Error

The goal is to compare how different baselines behave across short-term (1–3 days) and mid-term (4–7 days) forecasting horizons.

Baseline Forecasting Performance (H=1--7)

Metric	Model	H=1	H=2	H=3	H=4	H=5	H=6	H=7
MAPE	StemGNN	1.0676	1.8843	1.6147	1.9031	1.5162	1.7747	1.3762
	StemGNN_W	1.0776	1.8844	1.5896	1.8996	1.5347	1.7685	1.3735
	MTGNN	0.7657	2.2407	2.2425	2.2389	2.2324	2.2280	2.2311
	MTGNN_W	0.7510	2.2407	2.2425	2.2389	2.2324	2.2280	2.2311
	Last-Day	1.0149	0.9662	1.0130	1.0155	0.9909	0.9546	0.4254
	MovingAvg	0.9093	0.9071	0.9062	0.9067	0.9038	0.9042	0.9030
RMSE	StemGNN	2930.82	4513.43	4054.01	4722.15	3862.28	4419.07	3639.32
	StemGNN_W	2937.09	4526.22	3977.99	4734.25	3883.43	4410.45	3627.88
	MTGNN	2644.75	3564.70	3831.36	3815.54	3795.81	3795.63	3795.66
	MTGNN_W	2645.99	3564.70	3831.36	3815.54	3795.81	3795.63	3795.66
	Last-Day	4480.64	4580.82	4476.72	4486.12	4537.15	4469.88	2174.01
	MovingAvg	3089.70	3078.83	3090.44	3106.34	3125.29	3144.24	3165.23
MAE	StemGNN	1405.92	2914.68	2464.33	3013.20	2286.82	2773.10	2057.77
	StemGNN_W	1427.40	2916.38	2407.73	3010.59	2323.46	2761.42	2048.21
	MTGNN	1013.45	2545.51	2580.17	2572.93	2571.45	2571.03	2571.54
	MTGNN_W	1032.45	2545.51	2580.17	2572.93	2571.45	2571.03	2571.54
	Last-Day	1530.27	1594.86	1598.71	1596.00	1577.98	1507.35	458.15
	MovingAvg	1279.11	1270.16	1271.40	1274.76	1281.89	1289.30	1299.47

Modified StemGNN

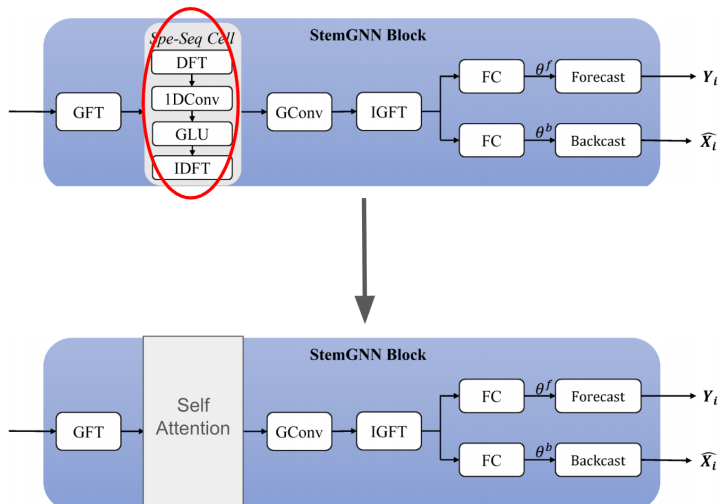


Figure: StemGNN Modification – 1

Modified StemGNN

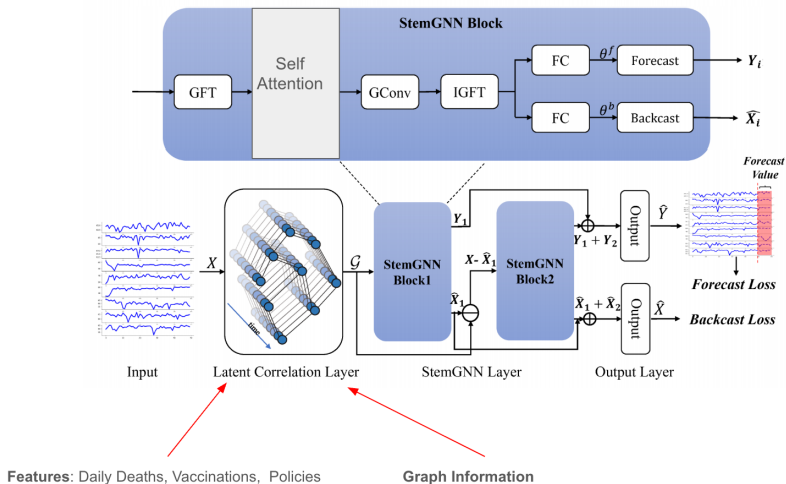


Figure: StemGNN Modification – 2

Hybrid GCN

- The idea here is similar to the MSST paper presented a couple of weeks ago
- We use the state-level graph to get its hidden representations, then concatenate it with the raw data and feed it into a GRU
- Two key changes:
 - We predict the *residuals of the moving average* not raw case counts
 - We add the predicted residuals to the moving average when calculating the loss
 - This model also predicts each variant separately instead of total cases

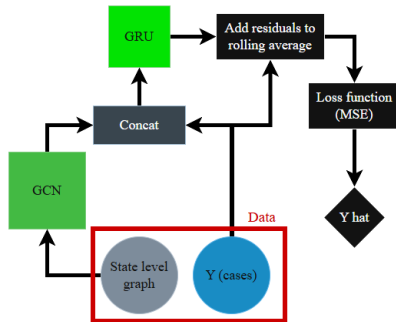
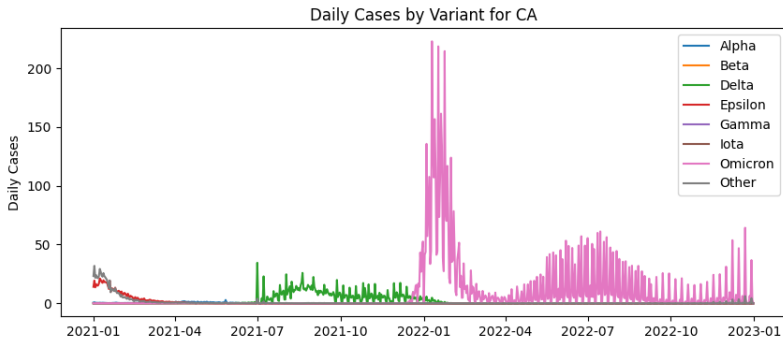


Figure: Hybrid Model structure

Hybrid GCN

- Why predict residuals?
 - The moving average baseline is already very strong, so by predicting only the residuals we take advantage of its strength
- Why predict each variant separately?
 - Each variant has its own patterns that are ignored when aggregated
- Other changes
 - The county-level graph and covariates were not used as it dramatically increased training time for zero/negative gain



DCGRNN: Diffusion Convolutional Graph RNN

- A deep learning architecture that combines **temporal dynamics** (RNN/GRU) with **spatial diffusion** (graph convolution).
- Leverages a diffusion process to propagate information across a graph.

Key Components

- **Graph:** Airport (0.33), Border (0.33), Highway (0.34)
- **RNN:** GRU with diffusion convolution
- **Features:** 12 covariates
- **Training:** 30 epochs, horizons $h = 1, \dots, 28$

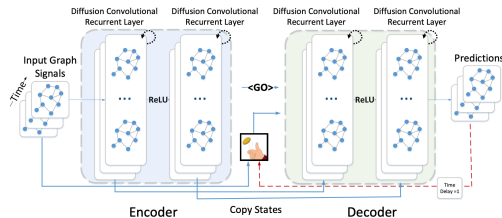


Figure: DCGRNN Architecture Diagram

Model Evaluation Results Overview

We evaluate all models on forecasting horizons from 1 to 7 days. The models are:

- **StemGNN_R**: our modified version of StemGNN with improved data alignment
- **Hybrid GCN**
- **DCGRNN**

For each model we report the same three metrics as before (MAPE, RMSE, MAE)

Overall Forecasting Performance of Models (H=1--7)

Metric	Model	H=1	H=2	H=3	H=4	H=5	H=6	H=7	Avg	% Under MA
MAPE	MovingAvg	0.9093	0.9071	0.9062	0.9067	0.9038	0.9042	0.9030	0.9058	-
	Hybrid GCN ¹	0.8345	0.9656	0.7946	0.9000	0.8915	0.7184	0.6512	0.8212	9.34%
	StemGNN_R	1.4584	1.7018	1.2760	1.0568	1.1516	1.8414	1.5832	1.438	-58.81%
	DCGRNN	1.4498	1.2862	1.3798	1.5830	1.6576	1.6529	1.7680	1.5396	-69.97%
RMSE	MovingAvg	3089.70	3078.83	3090.44	3106.34	3125.29	3144.24	3165.23	3113.63	-
	Hybrid GCN	2297.65	2713.47	2526.24	2457.98	2343.50	2237.08	2265.41	2411.11	22.56%
	StemGNN_R	3634.50	4229.53	3496.77	3309.49	3432.45	4630.62	4133.28	3838.09	-23.27%
	DCGRNN	2799.10	2901.81	3154.05	3319.53	3257.12	3483.43	3443.98	3194.15	-2.59%
MAE	MovingAvg	1279.11	1270.16	1271.40	1274.76	1281.89	1289.30	1299.47	1280.87	-
	Hybrid GCN	1018.97	1285.82	1089.05	1218.76	1158.94	846.266	818.543	1062.34	17.06%
	StemGNN_R	2121.67	2649.14	1851.14	1465.37	1654.35	2962.36	2496.03	2171.44	-69.53%
	DCGRNN	1315.03	1453.86	1591.14	1614.25	1544.03	1667.24	1658.4	1549.14	-17.32%

¹Values here are shown for total cases, aggregating the individual variant predictions

DCGRNN Performance Daily vs. Weekly Forecasting

Daily DCGRNN (1–28 days):

H	RMSE	MAE	MAPE
1	2799.1	1315.0	144.98%
2	2901.8	1453.9	128.62%
3	3154.0	1591.1	137.98%
4	3319.5	1614.2	158.30%
5	3257.1	1554.0	165.76%
6	3483.4	1667.2	165.29%
7	3444.0	1658.4	176.80%
14	3109.2	1523.0	138.63%
21	3768.1	1781.5	206.74%
28	3624.0	1713.8	172.44%

Weekly DCGRNN (1, 2, 4 weeks):

Horizon	MAE	RMSE	MAPE
1 week	99.7	172.9	26.81%
2 weeks	115.1	204.6	31.22%
4 weeks	119.6	211.4	33.04%

DCGRNN worked better for weekly:

- Weekly smoothing reduces noise → DCGRNN learns more stable patterns.
- Weekly MAE grows slowly (99.7→119.6) vs. daily (2799→3624).
- Weekly MAPE stays moderate (26–33%) vs. daily variability (130–200%).
- Aggregation enhances temporal structure and spatial diffusion cues.

Takeaways and Conclusion

- Overall, **Hybrid GCN** performed the best, with moderate improvements over the baseline
- Despite this, our experiments show that a simple moving average is already a decent predictor
 - This could be due to the nature of how disease spread, with even waves of cases not rising too quickly for MA to pickup on
 - This also could be a result of how cases are reported (every so often and not daily) - further work here should investigate how to best account for this
- It's also interesting that including county information and other covariates didn't help here
 - The graph structure might soak up most of the gains here, since transmissibility/density etc. would be (partially) captured by airport and highway networks

Questions?